EconSkills: Studying Skill Transfer and Retrieval for Web Agents on Live Economic Data

Yinzhu Quan, Zefang Liu

Paper version: arXiv 2609.19523v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *are each solved from scratch, and whatever an agent learns while solving the first is discarded before it reaches the second.*

Web agents revisit the same portals, yet **discard** the procedures learned on earlier successes.

Two questions, usually conflated

1. **Transfer.** Does a matched skill help?
2. **Selection.** Does a whole library keep it?
3. **Economics.** Series, units, vintages demand checks.

Evaluation setup

EconWebArena · 360 live | 50 skills · 37 portals | gpt-5-mini agent | gpt-5 extractor | 30-step budget | seeds 0, 1, 2

- **All online**; reward = gold value + authoritative final URL.
- Held out: 100 task variants; new instance values.

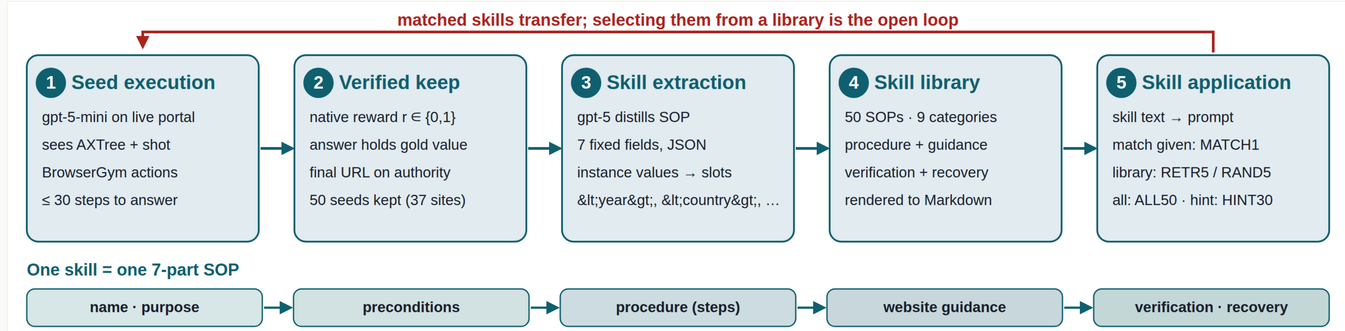
Our read (CUA Radar)

62	★★	online
score of 100	band	evidence level

- **Reports:** online eval on live sites; 3 seeds + std (controlled study); ablation; paired tests. **Not:** code; deployment variance; dollar cost.
- **Flags:** skill families overlap the benchmark (held-out variants mitigate); no external skill baselines; one backbone, one benchmark.
- Pattern K27. Authors' numbers, not rerun.

Method: verified trajectory → 7-part SOP → prompt

- **Sees / acts:** AXTree + screenshot + metadata in; BrowserGym actions (click, type, scroll, tabs) out.
- **Skill:** 7-field SOP; instance values become <year>, <country> placeholders.
- **Signal:** only runs that passed the native benchmark check become skills; injected as Markdown.



Our redraw (not the paper's figure): from a verified live run to a parameterized skill, then back into the prompt.

$$r_c(t) = 1 \iff \text{answer}_t \text{ holds gold } \wedge \text{ dom}(u_t) = \text{authority}$$

native EconWebArena reward kept for every condition

Key results: choosing from the library (live SR %)

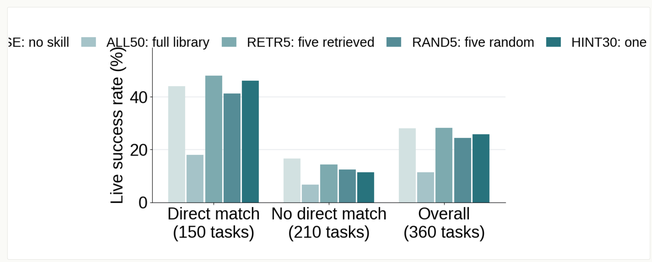


Table 4, redrawn: all 360 tasks, one run per task; direct-match split fixed by benchmark construction.

Match given, not selected: 41.0 → 49.3 (+8.3 pts, p = 0.0031)

SO WHAT → *RETR5 leads at 48.0% with a family match; ALL50 falls to 11.4% — context volume and retrieval decide deployment.*

Controlled transfer: what the prompt contains

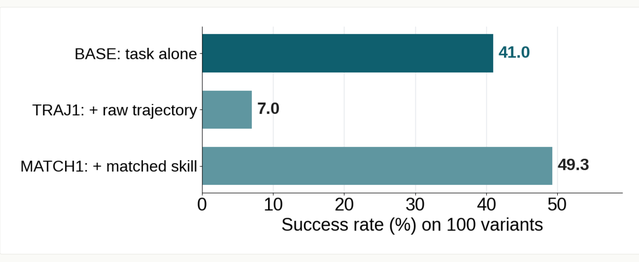


Table 3, redrawn: 100 held-out variants, 3 runs each (seeds 0, 1, 2); std 2.0 / 1.0 / 3.1.

SO WHAT → *Packaged SOP beats the raw-trajectory prompt 49.3 vs 7.0 — the paper notes the prompts also differ in length and format, so not abstraction alone. Mean steps 19.83 / 7.95 / 15.67: TRAJ1 ends early, it does not finish fast. On paired successes MATCH1 uses 2.37 fewer actions (median two fewer).*

Headline Numbers

49.3%
Matched-skill SR
vs 41.0% BASE no-skill, 3 runs

- **41.0%** BASE no skill
- **+8.3** p = 0.0031 vs BASE
- **28.3%** RETR5 overall (360)

The paper's own controls

CONTROL	NUMBERS	READING
TRAJ1 / BASE / MATCH1	7.0 / 41.0 / 49.3	SOP wins
Paired flips (300)	46 up / 21 down	McNemar p = 0.0031
Steps, both-success	2.37 fewer actions	sign test p = 0.0012
No-match hint (210)	16.7 → 11.4%	p = 0.043, hurts

Takeaway

A matched, verified SOP transfers (+8.3 pts) and shortens paired wins; end-to-end gains hinge on rejecting approximate candidates.

50 skills · 37 portals | 9 of 10 categories | ALL50: 11.4% overall | 1 backbone · 30 steps

Scan to read the full paper

arxiv.org/abs/2609.19523v2